高等量子力学　第 1 次作业

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题目一　设三维希尔伯特空间基矢为 $|a_1\rangle$、$|a_2\rangle$、$|a_3\rangle$。

(a) 写出 $|a_1\rangle$、$|a_3\rangle\langle a_2|$ 的矩阵表示。

(b) 求

$$|a_3\rangle\langle a_2| + |a_1\rangle\langle a_1| + |a_2\rangle\langle a_3| \tag{1}$$

的本征值和本征矢。

回答

(a)

$$|a_1\rangle \to \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad |a_3\rangle\langle a_2| \to \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}. \tag{2}$$

(b)

$$\hat{A} \equiv |a_3\rangle\langle a_2| + |a_1\rangle\langle a_1| + |a_2\rangle\langle a_3| \to \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \equiv A. \tag{3}$$

矩阵 A 的本征解为

$$\text{Eigenvals: } -1, 1, 1 \qquad \text{Eigenvecs: } \begin{pmatrix} 0 & -1 & 1 \end{pmatrix}^\top, \begin{pmatrix} 0 & 1 & 1 \end{pmatrix}^\top, \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}^\top \tag{4}$$

相应算符 $\hat{A}$ 的本征解为

$$\text{Eigenvals: } -1, 1, 1 \qquad \text{Eigenvecs: } \frac{1}{\sqrt{2}}\left(-|a_2\rangle + |a_3\rangle\right), \quad \frac{1}{\sqrt{2}}\left(|a_2\rangle + |a_3\rangle\right), \quad |a_1\rangle. \tag{5}$$

题目二　根据厄米共轭算符的定义，证明：

(a) $\left(|\psi\rangle\langle\phi|\right)^\dagger = |\phi\rangle\langle\psi|$；

(b) $\left(\hat{A}^\dagger\right)^\dagger = \hat{A}$；

(c) $\left(\hat{A}\hat{B}\right)^\dagger = \hat{B}^\dagger\hat{A}^\dagger$。

回答

(a) 由厄米共轭算符的定义，$\forall\, |\alpha\rangle, |\beta\rangle$,

$$\begin{gathered} \left\langle \alpha \middle| \left(|\psi\rangle\langle\phi|\right)^\dagger \middle| \beta \right\rangle = \langle\beta|\left(|\psi\rangle\langle\phi|\right)|\alpha\rangle^* = \langle\alpha|\phi\rangle\langle\psi|\beta\rangle = \langle\alpha|\left(|\phi\rangle\langle\psi|\right)|\beta\rangle, \\ \Longrightarrow \left(|\psi\rangle\langle\phi|\right)^\dagger = |\phi\rangle\langle\psi|. \end{gathered} \tag{6}$$

(b) 由厄米共轭算符的定义，$\forall |\alpha\rangle, |\beta\rangle$,

$$\begin{aligned} \left\langle \alpha \left| \left(A^\dagger\right)^\dagger \right| \beta \right\rangle = \left\langle \beta \left| A^\dagger \right| \alpha \right\rangle^* = (\langle \alpha | A | \beta \rangle^*)^* = \langle \alpha | A | \beta \rangle, \\ \Longrightarrow \left(A^\dagger\right)^\dagger = A. \end{aligned} \tag{7}$$

(c) 由厄米共轭算符的定义，$\forall |\alpha\rangle, |\beta\rangle$，一方面

$$\left\langle \alpha \left| (AB)^\dagger \right| \beta \right\rangle = \langle \beta | AB | \alpha \rangle^*, \tag{8}$$

另一方面

$$\left\langle \alpha \left| B^\dagger A^\dagger \right| \beta \right\rangle = \langle \alpha | B^\dagger \left(A^\dagger | \beta \rangle\right) = \langle \beta | AB | \alpha \rangle^*. \tag{9}$$

对比可知

$$(AB)^\dagger = B^\dagger A^\dagger. \tag{10}$$

题目三 导出坐标表象下，下列算符矩阵元的表达式：(a) $(\hat{x}\hat{p})^2$； (b) $\mathrm{e}^{\mathrm{i}a\hat{p}}$； (c) $\mathrm{e}^{\mathrm{i}p\hat{x}}$。

回答

(a) 利用 $\langle x | \hat{p}^n | x' \rangle = (-\mathrm{i}\hbar\partial_x)^n \delta(x - x')$, $[\hat{x}, \hat{p}] = \mathrm{i}\hbar$，有

$$\begin{aligned} (\hat{x}\hat{p})^2_{xx'} &= \left\langle x \left| \hat{x}\hat{p}\hat{x}\hat{p} \right| x' \right\rangle = x \left\langle x \left| \hat{p}\hat{x}\hat{p} \right| x' \right\rangle = x \left\langle x \left| (\hat{x}\hat{p} - \mathrm{i}\hbar)\hat{p} \right| x' \right\rangle \\ &= x^2 \left\langle x \left| \hat{p}^2 \right| x' \right\rangle - \mathrm{i}\hbar x \left\langle x \left| \hat{p} \right| x' \right\rangle = (-\mathrm{i}\hbar)^2 x^2 \frac{\partial^2 \delta(x - x')}{\partial x^2} + (-\mathrm{i}\hbar)^2 x \frac{\partial \delta(x - x')}{\partial x} \\ &= -\hbar^2 \left[x^2 \frac{\partial^2 \delta(x - x')}{\partial x^2} + x \frac{\partial \delta(x - x')}{\partial x} \right]. \end{aligned} \tag{11}$$

(b)

$$\begin{aligned} \left(\mathrm{e}^{\mathrm{i}a\hat{p}}\right)_{xx'} &= \left\langle x \left| \mathrm{e}^{\mathrm{i}a\hat{p}} \right| x' \right\rangle = \left\langle x \left| \sum_{n=0}^{\infty} \frac{(\mathrm{i}a\hat{p})^n}{n!} \right| x' \right\rangle = \sum_{n=0}^{\infty} \frac{1}{n!} (\mathrm{i}a)^n \langle x | \hat{p}^n | x' \rangle \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} (\mathrm{i}a)^n (-\mathrm{i}\hbar\partial_x)^n \delta(x - x') = \left[\sum_{n=0}^{\infty} \frac{1}{n!} (a\hbar\partial_x)^n \right] \delta(x - x') \\ &= \sum_{n=0}^{\infty} \frac{\delta^{(n)}(x - x')}{n!} (a\hbar)^n = \delta(x - x' + a\hbar). \end{aligned} \tag{12}$$

(c)

$$\begin{aligned} \left(\mathrm{e}^{\mathrm{i}p\hat{x}}\right)_{xx'} &= \left\langle x \left| \mathrm{e}^{\mathrm{i}p\hat{x}} \right| x' \right\rangle = \left\langle x \left| \sum_{n=0}^{\infty} \frac{(\mathrm{i}p\hat{x})^n}{n!} \right| x' \right\rangle = \left[\sum_{n=0}^{\infty} \frac{(\mathrm{i}px)^n}{n!} \right] \delta(x - x') \\ &= \mathrm{e}^{\mathrm{i}px} \delta(x - x'). \end{aligned} \tag{13}$$

题目四 证明：

(a) 厄米算符的本征值为实数；

(b) 反厄米算符的本征值为纯虚数；

(c) 幺正算符的本征值为模为 1 的复数。

回答

(a) 设厄米算符 $A = A^\dagger$ 的本征方程为 $A\left|a\right\rangle = a\left|a\right\rangle$，一方面有

$$\left\langle a\right|A\left|a\right\rangle = a, \qquad \left\langle a\right|\left.a\right\rangle = 1. \tag{14}$$

另一方面，

$$\left\langle a\right|A\left|a\right\rangle = \left\langle a\right|A^\dagger\left|a\right\rangle = \left\langle a\right|A\left|a\right\rangle^* = a^*, \tag{15}$$

于是 $a = a^* \Longrightarrow a \in \mathbb{R}$.

(b) 设反厄米算符 $A = -A^\dagger$ 的本征方程为 $A\left|a\right\rangle = a\left|a\right\rangle$，一方面有

$$\left\langle a\right|A\left|a\right\rangle = a, \tag{16}$$

另一方面，

$$\left\langle a\right|A\left|a\right\rangle = \left\langle a\right|-A^\dagger\left|a\right\rangle = -\left\langle a\right|A\left|a\right\rangle^* = -a^*, \tag{17}$$

于是 $a = -a^* \Longrightarrow a \in \mathrm{i}\mathbb{R}$ 。

(c) 设幺正算符 U 的本征方程为 $U\left|u\right\rangle = u\left|u\right\rangle$，则

$$1 = \left\langle u\right|\left.u\right\rangle = \left\langle u\right|U^\dagger U\left|u\right\rangle = \left|u\right|^2\left\langle u\right|\left.u\right\rangle = \left|u\right|^2 \Longrightarrow \left|u\right| = 1. \tag{18}$$

题目五 设

$$F_{x,x'} = -\mathrm{i}x^2\frac{\partial}{\partial x}\delta(x - x'), \tag{19}$$

求 $F_{p,p'}$。

回答

$$\begin{aligned}
F_{p,p'} &= \left\langle p\right|F\left|p'\right\rangle = \int \mathrm{d}x\mathrm{d}x'\left\langle p\right|\left.x\right\rangle\left\langle x\right|F\left|x'\right\rangle\left\langle x'\right|\left.p'\right\rangle \\
&= \int \mathrm{d}x\mathrm{d}x'\left(\frac{1}{\sqrt{2\pi\hbar}}\mathrm{e}^{-\mathrm{i}px/\hbar}\right)\left[-\mathrm{i}x^2\frac{\partial}{\partial x}\delta(x - x')\right]\left(\frac{1}{\sqrt{2\pi\hbar}}\mathrm{e}^{\mathrm{i}p'x'/\hbar}\right) \\
&= \frac{-\mathrm{i}}{2\pi\hbar}\int \mathrm{d}x x^2\mathrm{e}^{-\mathrm{i}px/\hbar}\frac{\partial}{\partial x}\int \mathrm{d}x'\delta(x - x')\mathrm{e}^{\mathrm{i}p'x'/\hbar} = \frac{p'}{2\pi\hbar^2}\int \mathrm{d}x x^2\mathrm{e}^{\mathrm{i}(p'-p)x/\hbar} \\
&= \frac{p'}{2\pi\hbar^2}\left(\frac{\hbar}{\mathrm{i}}\right)^2\int \mathrm{d}x\frac{\partial^2}{\partial\left(p' - p\right)^2}\mathrm{e}^{\mathrm{i}(p'-p)x/\hbar} = \frac{-p'}{2\pi}\frac{\mathrm{d}^2}{\mathrm{d}\left(p' - p\right)^2}\int \mathrm{d}x\mathrm{e}^{\mathrm{i}(p'-p)x/\hbar} \\
&= \frac{-p'}{2\pi}\frac{\mathrm{d}^2}{\mathrm{d}\left(p' - p\right)^2}\left[2\pi\delta\left(\frac{p' - p}{\hbar}\right)\right] = -\hbar p'\frac{\mathrm{d}^2}{\mathrm{d}\left(p' - p\right)^2}\delta(p' - p) \\
&= -\hbar p'\frac{\partial^2}{\partial p'^2}\delta(p' - p).
\end{aligned} \tag{20}$$
